

Exercises 7

Exercise 1 (Fixed points of real functions). Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$f(x) = x^2 - x - 3 .$$

Describe the set $F_f \subseteq \mathbb{R}$, the set of fixed points of f . Does f admit a least, or respectively greatest, fixed point?

Solution: *Every fixed point x of f satisfies the equation*

$$f(x) = x$$

and thus

$$\begin{aligned} x^2 - x - 3 &= x \\ \iff x^2 - 2x - 3 &= 0 \\ \iff (x - 3)(x + 1) &= 0 . \end{aligned}$$

The only real numbers satisfying this equation are $x = 3$ and $x = -1$. Therefore, $F_f = \{-1, 3\}$.

F_f is finite and non-empty, so it has both a least and a greatest element (-1 and 3). $\diamond$

Exercise 2 (Iterating sp and wp). This question is inspired by relationships between state sets in a transition system.

Let S be a non-empty set of states and $r \subseteq S \times S$ a relation. As usual, define

- for every $P \subseteq S$, $sp(P, r) = \{x' \mid \exists x \in P. (x, x') \in r\}$
- for every $Q \subseteq S$, $wp(r, Q) = \{x \mid \forall x'. (x, x') \in r \rightarrow x' \in Q\}$

Let $2^S = \{X \mid X \subseteq S\}$ be the set of all sets of states. Let $Init \subseteq S$ and $Good \subseteq S$. Define:

- $F : 2^S \rightarrow 2^S$ as $F(X) = Init \cup sp(X, r)$
- $B : 2^S \rightarrow 2^S$ as $B(Y) = Good \cap wp(r, Y)$
- $l = \bigcup_{i \geq 0} F^i(\emptyset)$
- $h = \bigcap_{i \geq 0} B^i(S)$

Consider the partial order $(2^S, \subseteq)$. For each of the following properties on this order, determine if it holds or not. Provide a proof or a counterexample in each case.

1. F is monotonic

Solution: True. For any $X \subseteq X'$, we have $sp(X, r) \subseteq sp(X', r)$, and thus $F(X) \subseteq F(X')$. $\diamond$

2. B is monotonic

Solution: True. For any $Y \subseteq Y'$, we have $wp(r, Y) \subseteq wp(r, Y')$, and thus $B(Y) \subseteq B(Y')$ by monotonicity (on the right) of $\cap$. $\diamond$

3. l is the least fixed point of F .

Solution: True. We will show that F is ω -continuous to prove that l is a fixed point. F is ω -continuous if for every increasing sequence $X_0 \subseteq X_1 \subseteq \dots$ we have

$$F\left(\bigcup_{i \geq 0} X_i\right) = \bigcup_{i \geq 0} F(X_i)$$

We will prove inclusions in both directions.

(a) **Forward**, we need to show that

$$F\left(\bigcup_{i \geq 0} X_i\right) \subseteq \bigcup_{i \geq 0} F(X_i).$$

We will show that for any $x \in F\left(\bigcup_{i \geq 0} X_i\right)$, we have $x \in \bigcup_{i \geq 0} F(X_i)$.

If $x \in F\left(\bigcup_{i \geq 0} X_i\right)$, then either $x \in \text{Init}$ or $x \in sp\left(\bigcup_{i \geq 0} X_i, r\right)$ (or both). We split cases:

- If $x \in \text{Init}$, then for every $i \geq 0$ we have $x \in F(X_i)$, so $x \in \bigcup_{i \geq 0} F(X_i)$.
- If $x \in sp\left(\bigcup_{i \geq 0} X_i, r\right)$, then there exists $y \in \bigcup_{i \geq 0} X_i$ such that $(y, x) \in r$. Since $y \in \bigcup_{i \geq 0} X_i$, there exists $j \geq 0$ such that $y \in X_j$. It follows that $x \in sp(X_j, r)$, and thus $x \in F(X_j)$. Therefore, $x \in \bigcup_{i \geq 0} F(X_i)$.

Since in both cases we have $x \in \bigcup_{i \geq 0} F(X_i)$, the desired inclusion holds.

(b) **Backward**, we need to show that

$$\bigcup_{i \geq 0} F(X_i) \subseteq F \left(\bigcup_{i \geq 0} X_i \right) .$$

By definition of $\bigcup$, for every $i \geq 0$ we have $X_i \subseteq \bigcup_{i \geq 0} X_i$. By monotonicity of F , we have $F(X_i) \subseteq F \left(\bigcup_{i \geq 0} X_i \right)$ for each i . It follows that $\bigcup_{i \geq 0} F(X_i) \subseteq F \left(\bigcup_{i \geq 0} X_i \right)$.

We can finally show that l is the least fixed point of F . We need to show that $F(l) = l$. We have

$$\begin{aligned} & F(l) \\ &= F \left(\bigcup_{i \geq 0} F^i(\emptyset) \right) && \text{(definition of } l) \\ &= \bigcup_{i \geq 0} F(F^i(\emptyset)) && (\omega\text{-continuity of } F) \\ &= \bigcup_{i > 0} F^i(\emptyset) \\ &= \emptyset \cup \bigcup_{i > 0} F^i(\emptyset) \\ &= F^0(\emptyset) \cup \bigcup_{i > 0} F^i(\emptyset) \\ &= \bigcup_{i \geq 0} F^i(\emptyset) \\ &= l . \end{aligned}$$

Thus, l is a fixed point of F . To show that it is the least such fixed point, we will show that any fixed point k of F upper bounds the sequence $(F^i(\emptyset))_{i \geq 0}$.

We proceed by induction. For the base case, $F^0(\emptyset) = \emptyset \subseteq k$. For the inductive step, assume the claim holds for some $i \geq 0$. Then we have

$$\begin{aligned} & F^i(\emptyset) \subseteq k && \text{(I.H.)} \\ \implies & F^{i+1}(\emptyset) \subseteq F(k) && \text{(monotonicity of } F) \\ \implies & F^{i+1}(\emptyset) \subseteq k . && (k \text{ is a fixed point of } F) \end{aligned}$$

Since k upper bounds the sequence, it also upper bounds its supremum. Thus, $l \subseteq k$ for any fixed point k of F , and so l is the least fixed point of F . $\diamond$

4. h is the greatest fixed point of B .

Solution: True. This can be shown directly using the $\supseteq$ instead of $\subseteq$ order, or as follows. From lectures and exercises (set 4), we know that

$$wp(r, X) = sp(X^c, r^{-1})^c$$

for any $X \subseteq S$, where $X^c = S \setminus X$.

Define an operation dual to B , $B_1 : 2^S \rightarrow 2^S$ as

$$\begin{aligned} B_1(Z) &= B(Z^c)^c \\ &= \left(Good \cap wp(r, Z^c) \right)^c \\ &= Good^c \cup wp(r, Z^c)^c \\ &= Good^c \cup sp(Z, r^{-1}) \\ &= Bad \cup sp(Z, r^{-1}) \end{aligned}$$

where $Bad = Good^c$. Note that B_1 has the same form as F , but with r^{-1} as relation and Bad in place of $Init$.

By the same argument as the previous part, B_1 is ω -continuous, and its least fixed point is given by

$$h_1 = \bigcup_{i \geq 0} B_1^i(\emptyset) .$$

This can be seen as the set of states reachable from Bad (error states) going backwards. By induction on i , for $i \geq 0 \in \mathbb{N}$, we can show that

$$B_1^i(\emptyset) = (B^i(S))^c .$$

For the base case, $S^c = \emptyset$. For the inductive step, assume the claim holds for some $i \geq 0$. Then we have

$$\begin{aligned} B_1^{i+1}(\emptyset) &= B_1(B_1^i(\emptyset)) \\ &= B_1\left((B^i(S))^c\right) && \text{(I.H.)} \\ &= (B(B^i(S)))^c && \text{(definition of } B_1) \\ &= (B^{i+1}(S))^c . \end{aligned}$$

In particular, we have that $h_1 = h^c$. Since h_1 is the least fixed point of B_1 , it follows that h is the greatest fixed point of B (complementary chains as above). $\diamond$

5. $l \subseteq h$.

Solution: **Not necessarily.** In fact, this condition holds precisely when *Good* is an invariant of a transition system with the initial set *Init* and a transition relation *r*. *l* represents reachable states starting from *Init* and making steps using the transition relation *r*. On the other hand, *h* represents states that are guaranteed to have their successors inside the set *Good*. As a counterexample, let $S = \{0, 1\}$, $Init = Good = \{0\}$ and $r = \{(0, 1)\}$. Then $l = \{0, 1\}$. Computation shows that $h = \emptyset$ because $Good \cap wp(r, Good) = \{0\} \cap \{1\} = \emptyset$. $\diamond$

6. $l \subseteq Good$ implies $Init \subseteq h$

Solution: **True.** Both conditions are stating that reachable states remain inside *Good*, that is, that there is no finite sequence $s_0, s_1, \dots, s_n$ such that $n \geq 0$, $s_0 \in Init$, $s_n \in Bad$ and, for all $0 \leq i < n$, $(s_i, s_{i+1}) \in r$:

$$\begin{aligned}
& l \subseteq Good \\
& \iff \bigcup_{i \geq 0} F^i(\emptyset) \subseteq Good \\
& \iff \forall i \geq 0. F^i(\emptyset) \subseteq Good \\
& \iff \forall i \geq 0. F^i(\emptyset) \cap Bad = \emptyset \\
& \iff \forall i \geq 0. r^i[Init] \cap Bad = \emptyset
\end{aligned}$$

whereas

$$\begin{aligned}
& Init \subseteq h \\
& \iff Init \subseteq \bigcap_{i \geq 0} B^i(S) \\
& \iff \forall i \geq 0. Init \subseteq B^i(S) \\
& \iff \forall i \geq 0. Init \subseteq (S \setminus B_1^i(\emptyset)) \\
& \iff \forall i \geq 0. Init \cap B_1^i(\emptyset) = \emptyset \\
& \iff \forall i \geq 0. Init \cap (r^{-1})^i[Bad] = \emptyset
\end{aligned}$$

$\diamond$

7. $Init \subseteq h$ implies $l \subseteq Good$

Solution: **True.** See the explanation for the previous case. $\diamond$

Exercise 3 (Fixed points in lattices). Consider a complete lattice with the set of elements L and the ordering relation $\sqsubseteq: L \times L$. Recall that in a

complete lattice, every set $X \subseteq L$ has a least upper bound and a greatest lower bound in L .

Let $G : L \rightarrow L$ be a monotonic function with respect to $\sqsubseteq$. Let

$$Fix = \{x \in L \mid G(x) = x\}$$

be the set of all fixed points. Let $x, y \in Fix$ be two fixed points. Prove or disprove:

1. $x \sqcup y \sqsubseteq G(x \sqcup y)$

Solution: True. We have $x \sqsubseteq x \sqcup y$ (because $x \sqcup y$ is an upper bound on $\{x, y\}$). By monotonicity of G , then $G(x) \sqsubseteq G(x \sqcup y)$. Analogously we obtain $G(y) \sqsubseteq G(x \sqcup y)$. This means that $G(x \sqcup y)$ is one upper bound on the set $\{G(x), G(y)\}$. Thus, the least among the upper bounds is smaller than it:

$$G(x) \sqcup G(y) \sqsubseteq G(x \sqcup y)$$

We have $G(x) = x$ and $G(y) = y$ because $x, y \in Fix$, so the desired property holds. $\diamond$

2. $G(x \sqcup y) \sqsubseteq x \sqcup y$

Solution: False. The property would hold if G distributed over $\sqcup$, so we need to find an example where this is not the case. There are many counterexamples, but let us take inspiration from the counter-example to union-distributivity of contexts E in the lectures.

Take L to be the set of all relations on the set $S = \{1, 2, 3\}$ with $\sqcup$ being $\cup$ and define $G(r) = \Delta_S \cup (r \circ r)$. Then any reflexive and transitive relation is a fixed point of G , including

$$r_1 = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$$

$$r_2 = \{(1, 1), (2, 2), (3, 3), (2, 3)\}$$

We gave $r_1 \cup r_2 = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$, which is not transitive. $G(r_1 \cup r_2)$ contains in particular $(1, 3)$, so $r_1 \cup r_2$ is not a fixed point of G . $\diamond$

3. $x \sqcup y \in Fix$

Solution: False. By definition, $x \sqcup y \in Fix$ means $G(x \sqcup y) = x \sqcup y$ and this implies $G(x \sqcup y) \sqsubseteq x \sqcup y$, which we have shown to be false in general in the previous part. $\diamond$

4. Let $B = \{b \in \text{Fix} \mid x \sqsubseteq b \wedge y \sqsubseteq b\}$. Then B has a least element, that is, there exists an element $z \in B$ such that $\forall b \in B. z \sqsubseteq b$. (Possibly difficult)

Solution: True. Note that such a least element will not generally be the least upper bound of x and y in L (which is $x \sqcup y$). However, it is their least upper bound in the lattice of fixed points, and we do have that $x \sqcup y \sqsubseteq z$ in the original lattice.

First, we define a superset of B , containing prefixed points satisfying the same criteria:

$$P = \{p \in \text{Prefix} \mid x \sqsubseteq p \wedge y \sqsubseteq p\}$$

where $\text{Prefix} = \{p \in L \mid G(p) \sqsubseteq p\}$ is the set of all prefixed points.

Since $(L, \sqsubseteq)$ is a complete lattice, $P \subseteq L$ has a greatest lower bound in L . Let us note this element by z . We will show that $z \in B$ and that z is a lower bound for B .

We need to show (in order) that z is a lower bound for B , that $x \sqsubseteq z$ and $y \sqsubseteq z$, and that $z \in \text{Fix}$.

- (a) z lower bounds B : z is a lower bound on P by definition. Since $B \subseteq P$, z is also a lower bound on B .
- (b) $x \sqsubseteq z$ and $y \sqsubseteq z$: note that $x \sqcup y$ is a lower bound on P , as any $b \in P$ upper bounds $\{x, y\}$, and thus must be greater-or-equal to their least upper bound.

Then, since z is the greatest lower bound on P , we have $x \sqcup y \sqsubseteq z$, which implies $x \sqsubseteq z$ and $y \sqsubseteq z$.

- (c) $z \in \text{Fix}$: we need to show that $G(z) = z$. We will show first that $G(z) \sqsubseteq z$ and then that $z \sqsubseteq G(z)$.

We know that for an arbitrary $b \in P$:

$$\begin{aligned} z &\sqsubseteq b && (z \text{ is glb of } P) \\ \implies G(z) &\sqsubseteq G(b) && (\text{monotonicity of } G) \\ \implies G(z) &\sqsubseteq b && (b \in \text{Prefix}) \end{aligned}$$

and thus $G(z)$ is a lower bound on P . Since z is the greatest lower bound on P , we have $G(z) \sqsubseteq z$.

We know that $x \sqsubseteq z$. By monotonicity of G , and the fact that $x \in \text{Fix}$, we have that $x \sqsubseteq G(z)$. Similarly for y , we get $y \sqsubseteq G(z)$.

From $G(z) \sqsubseteq z$ and monotonicity of G , we have $G(G(z)) \sqsubseteq G(z)$, so $G(z) \in \text{Prefix}$. Finally, we have $G(z) \in P$.

Since z is a lower bound on P , we have $z \sqsubseteq G(z)$. This proves that $G(z) = z$, so $z \in \text{Fix}$.

Given that $z \in \text{Fix}$, $x \sqsubseteq z$ and $y \sqsubseteq z$, we have $z \in B$. Since we know that z is a lower bound on B , we have shown that z is the least element of B .

◇

Exercise 4 (Abstract interpretation). Consider a program on two integer variables, that is the set of sets of states, i.e. the concrete domain, is $C = 2^{\mathbb{Z} \times \mathbb{Z}}$. We want to represent these states with an abstract domain A such that a state is represented by exact options for the first variable, and only as an interval for the second variable.

1. Express the lattice A corresponding to this domain.

Solution: Note by $\mathcal{I}(\mathbb{Z})$ the set of integer intervals. A can be represented as the set of functions $\mathbb{Z} \rightarrow \mathcal{I}(\mathbb{Z})$. Note that this is not the same as $2^{\mathbb{Z}} \times \mathcal{I}(\mathbb{Z})$.

We will illustrate why this is the case. A concrete state is a set of pairs (x, y) where x and y are integer values. Two possible examples are:

$$\begin{aligned} S_1 &= \{(1, 1), (1, 3), (2, 2)\} \\ S_2 &= \{(1, 1), (1, 2), (1, 3), (2, 2)\} \end{aligned}$$

First, note that both these sets are isomorphic to functions in $\mathbb{Z} \rightarrow 2^{\mathbb{Z}}$:

$$\begin{aligned} S_1 &\simeq f_1 = \{1 \mapsto \{1, 3\}, 2 \mapsto \{2\}\} \\ S_2 &\simeq f_2 = \{1 \mapsto \{1, 2, 3\}, 2 \mapsto \{2\}\} \end{aligned}$$

and more generally

$$\begin{aligned} &2^{\mathbb{Z} \times \mathbb{Z}} \\ &\simeq (\mathbb{Z} \times \mathbb{Z}) \rightarrow 2 \\ &\simeq \mathbb{Z} \rightarrow (\mathbb{Z} \rightarrow 2) \\ &\simeq \mathbb{Z} \rightarrow 2^{\mathbb{Z}} \end{aligned}$$

For the abstract domain, f_1 (resp. S_1) is not a valid abstract state set, as $f_1(1) = \{1, 3\}$ is not an interval. On the other hand, f_2 (resp. S_2) is a valid abstract state, as $f_2(1) = \{1, 2, 3\}$ is an interval $[1, 3] \in \mathcal{I}(\mathbb{Z})$.

Thus, in analogy to $C \simeq \mathbb{Z} \rightarrow 2^{\mathbb{Z}}$, we have $A \simeq \mathbb{Z} \rightarrow \mathcal{I}(\mathbb{Z})$, wherein the second variable is restricted to interval sets. For the rest of this exercise, we will identify A with $\mathbb{Z} \rightarrow \mathcal{I}(\mathbb{Z})$.

We need to define a lattice structure on A . We define the meet and join operations $\sqcap$ and $\sqcup$ on A .

Note by $\sqcup_{\mathcal{I}(\mathbb{Z})}$ and $\sqcap_{\mathcal{I}(\mathbb{Z})}$ the standard join and meet operations on integer intervals. We can define, for $f, g \in A$:

$$(f \sqcup g)(a) = f(a) \sqcup_{\mathcal{I}(\mathbb{Z})} g(a)$$

And similarly for $\sqcap$. From the fact that intervals form a lattice, it easily follows $(A, \sqcup, \sqcap)$ is a lattice. $\diamond$

2. Give expressions for monotonic functions $\alpha : C \rightarrow A$ and $\gamma : A \rightarrow C$ such that they form a (non-trivial) Galois connection.

Solution: Define α as

$$\alpha(s)(x) = \mathcal{I}(\{y \mid (x, y) \in s\})$$

where $\mathcal{I}(X) = [\min(X), \max(X)]$ for non-empty sets $X \subseteq \mathbb{Z}$, and $\mathcal{I}(\emptyset) = [0, -1]$ (any empty interval suffices).

$$\gamma(f) = \{(x, y) \mid y \in f(x)\}.$$

It is slightly tedious but not difficult to check that those two functions satisfy the conditions

$$\alpha(s) \sqsubseteq f \iff s \subseteq \gamma(f)$$

for all $s \in C$ and $f \in A$. $\diamond$

Exercise 5 (Program analysis). Consider a program that manipulates two integer variables x and y . Consider any assignment of the form $x := e$, where e is a (normalized) linear combination of integer variables, for example

$$x := 2 * x - 5 * y .$$

Consider an abstract analysis with independent intervals for both variables. Describe an algorithm that, given the expression e , and intervals for x (noted $[a_x, b_x]$) and y (noted $[a_y, b_y]$) computes the new interval $[a, b]$ for x after the assignment statement.

Provide an algorithm that takes as input e , and the intervals for x and y as below, and produces the new interval $[a, b]$ for x after the assignment statement. You may disregard integer overflows and underflows.

```

1 import math.*
2
3 // represents  $a*x + b*y + c$ 
4 case class LinearExpr(a: Int, b: Int, c: Int)
5 case class Interval(min: Int, max: Int)
6
7 //  $x := e$  under given intervals for  $x$  and  $y$ 
8 def nextInterval(
9     e: LinearExpr,
10    x: Interval,
11    y: Interval
12 ): Interval =
13     // your code here
14     ???

```

Solution:

```

1 import math.*
2
3 // represents  $a*x + b*y + c$ 
4 case class LinearExpr(a: Int, b: Int, c: Int)
5 case class Interval(min: Int, max: Int)
6
7 //  $x := e$  under given intervals for  $x$  and  $y$ 
8 def nextInterval(
9     e: LinearExpr,
10    x: Interval,
11    y: Interval
12 ): Interval =
13     // multiply  $x$ 
14     val nax = min(e.a * x.min, e.a * x.max)
15     val nbx = max(e.a * x.min, e.a * x.max)
16
17     // multiply  $y$ 
18     val nay = min(e.b * y.min, e.b * y.max)
19     val nby = max(e.b * y.min, e.b * y.max)
20
21     // compute
22     Interval(
23         min = nax + nay + e.c,
24         max = nbx + nby + e.c
25     )

```

◇